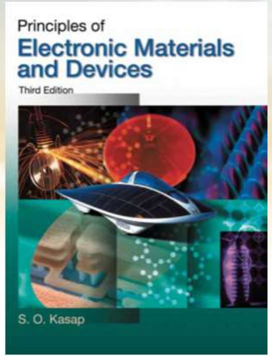
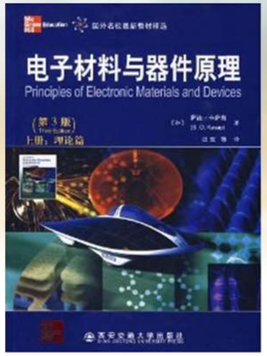
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》
Solid-State Electronics





Ch4-4
Modern
Theory of Solids

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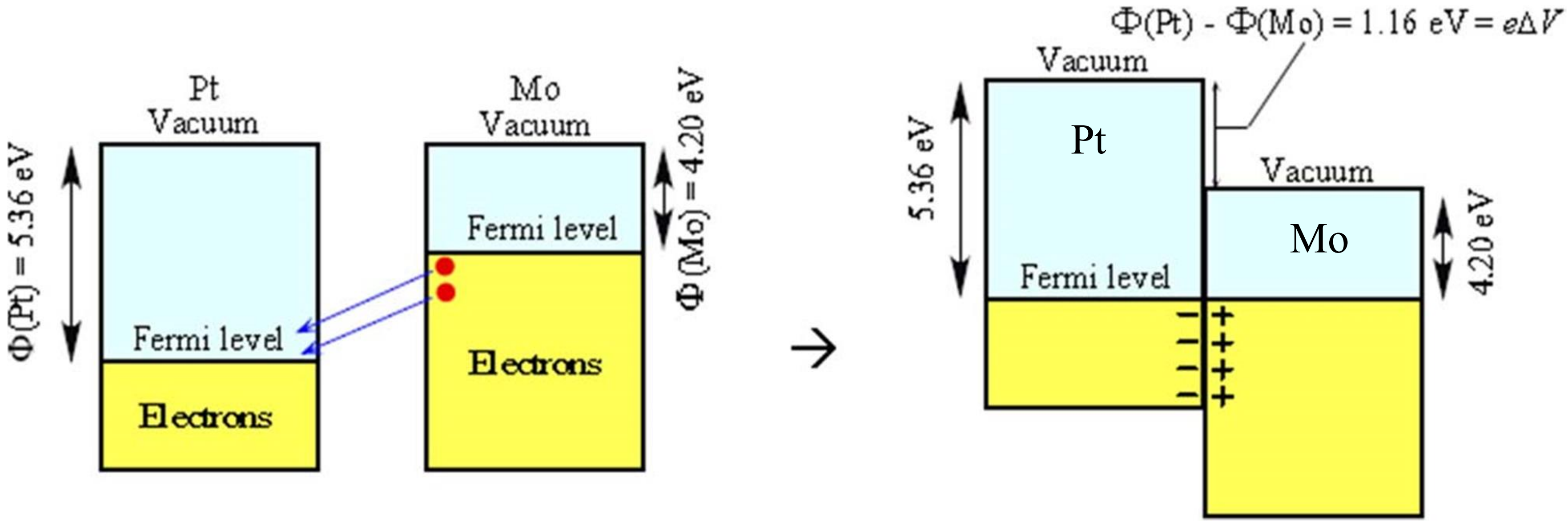
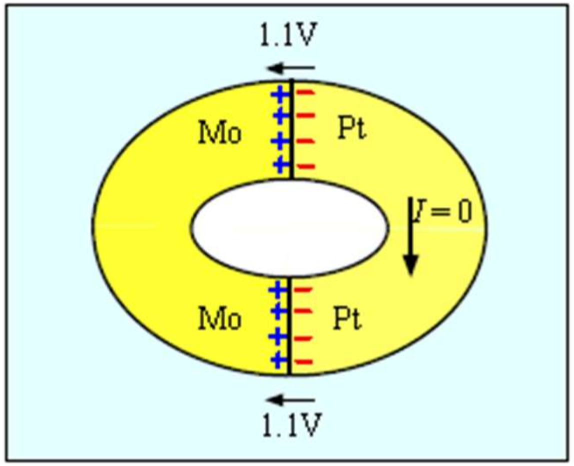
Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

Metal-Metal Contact



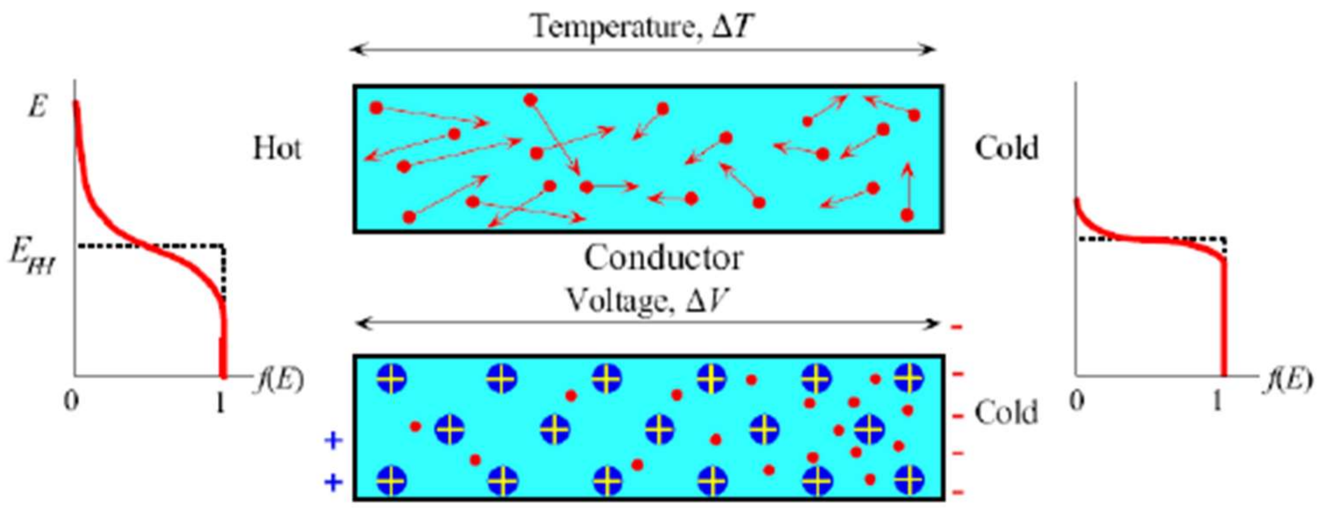


The contact voltage can do work?

However, there is **no current** when a closed circuit is formed from two different metals, even though there is a contact potential at each side.

This is because another junction with a contact voltage that is equal and opposite to that of the first junction.

Seebeck Effect



Seebeck Effect: the potential difference across a piece of metal due to a temperature difference. The magnitude of this effect is gauged by Seebeck coefficient.

$$S = \frac{dV}{dT}$$

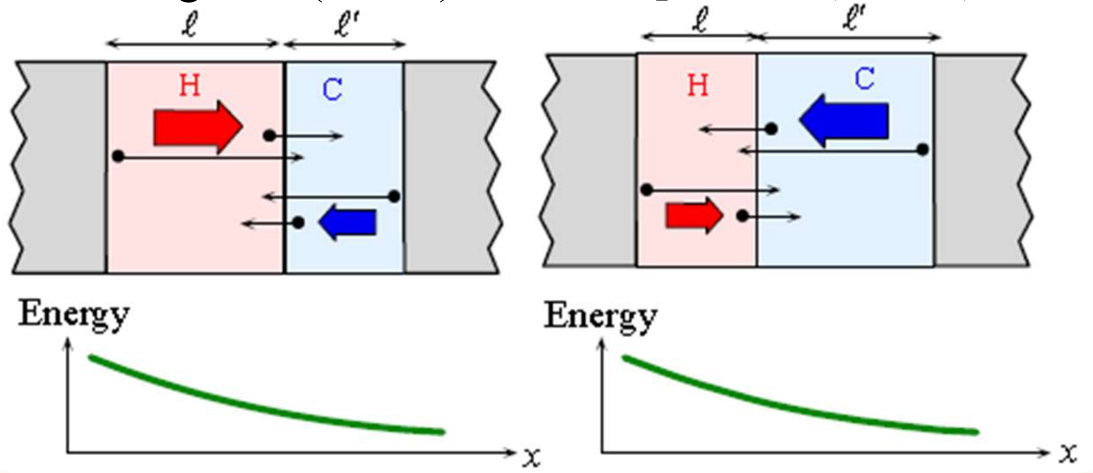
Sign of *S* represents the potential of the cold side with respect to the hot side.

Seebeck Effect

Consider two neighboring region H (hot) and C (cold) with widths corresponding to the mean free paths *l* and *l’* in H and C

Roughly speaking, with no ‘e’ scattering, net diffusion from H to C is $\propto \frac{1}{2} n (l - l')$

S negative (i.e. Al) *S* positive (i.e. Cu)



- *S* negative: free path *l* increases with *T*
- *S* positive: free path *l* decreases with *T*
- *l* affected by scattering such as lattice vibrations, impurities and crystal defects, etc.

Seebeck Effect

$S = S(T)$ → A material property that depends on temperature.

Given the $S(T)$ for a material, we can have the voltage difference between two points where temperature are T_0 and T.

$$\boxed{\Delta V = \int_{T_0}^{T} S dT}$$

The Seebeck coefficient for many metals is given by the Mott-Jones Equation

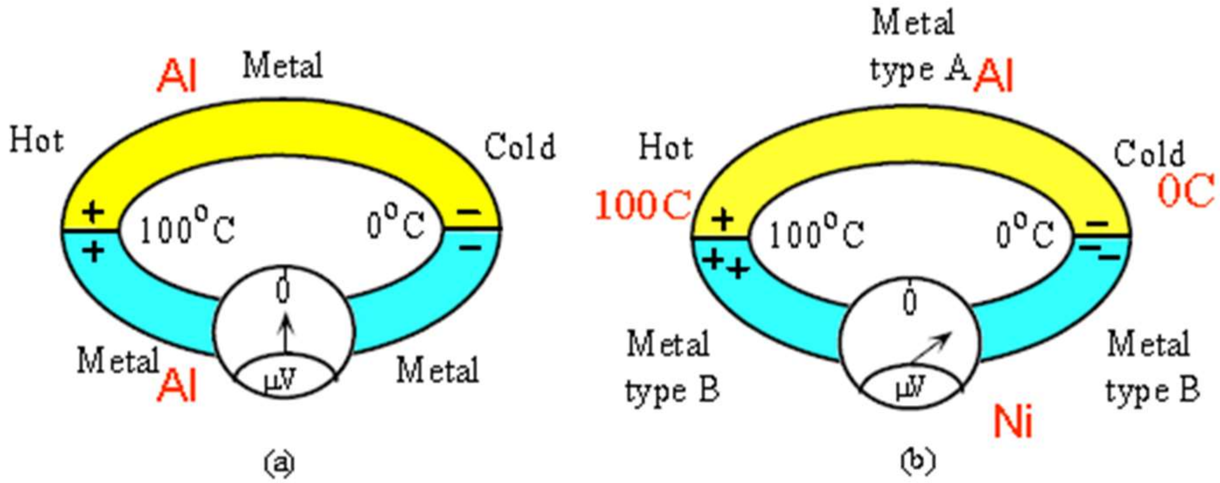
$$S \approx -\frac{\pi^2 k^2 T}{3eE_{F0}} x$$

x: numerical constant

Table 4.3 Seebeck coefficients of selected metals (from various sources)

Metal	S at 0 °C (μV K^{-1})	S at 27 °C (μV K^{-1})	E_F (eV)	x
Al	−1.6	−1.8	11.6	2.78
Au	+1.79	+1.94	5.5	−1.48
Cu	+1.70	+1.84	7.0	−1.79
K		−12.5	2.0	3.8
Li	+14		4.7	−9.7
Mg	−1.3		7.1	1.38
Na		−5	3.1	2.2
Pd	−9.00	−9.99		
Pt	−4.45	−5.28		

Thermocouple



If uses two different metals with one junction maintained at a reference temperature T_0 and the other used to sense the temperature T. The voltage across each metal element depends on its Seebeck coefficient.

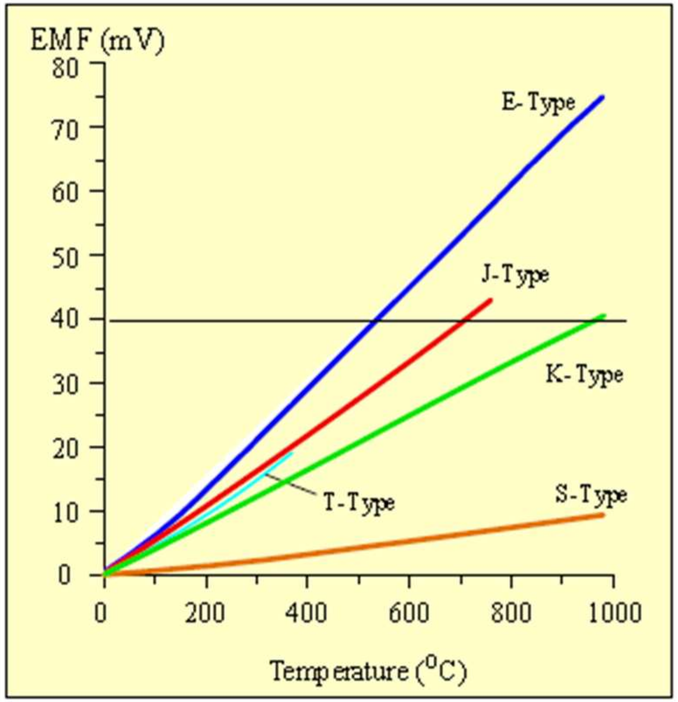
Electromotive force (emf) between two wires

$$V_{AB} = \int_{T_0}^{T} (S_A - S_B)\, dT = \int_{T_0}^{T} S_{AB}\, dT$$

Thermocouple

Table 4.4 Thermoelectric emf for metals at 100 and 200 °C with respect to Pt and the reference junction at 0 °C

	emf (mV)	
Material	**At 100 °C**	**At 200 °C**
Copper, Cu	0.76	1.83
Aluminum, Al	0.42	1.06
Nickel, Ni	−1.48	−3.10
Palladium, Pd	−0.57	−1.23
Platinum, Pt	0	0
Silver, Ag	0.74	1.77
Alumel	−1.29	−2.17
Chromel	2.81	5.96
Constantan	−3.51	−7.45
Iron, Fe	1.89	3.54
90% Pt–10% Rh (platinum-rhodium)	0.643	1.44



thermocouple equation $$V_{AB} = a\Delta T + B(\Delta T)^2$$

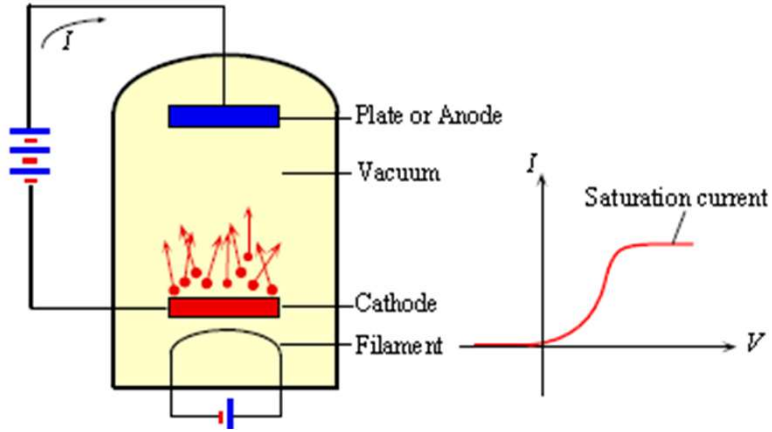
Thermocouple

Thermocouples are widely used to measure the temperature.

LEFT: A thermocouple pair embedded in a stainless steel sheath-probe. The thermocouple junction inside the probe is in thermal contact with the probe tip, and, electrically insulated from the probe metal.

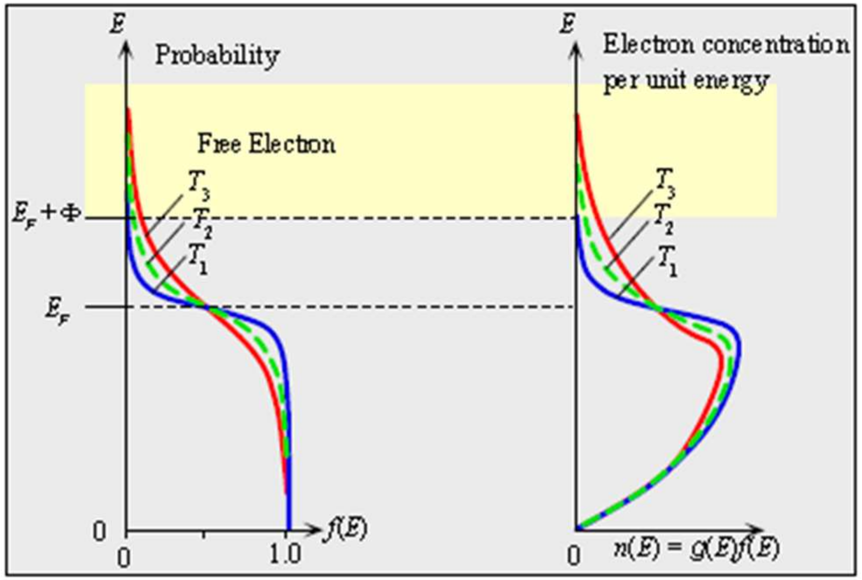
SOURCE: Courtesy of Omega

Thermionic Emission and Vacuum Tubes



$$E = \frac{1}{2}m_e v_x^2 + \frac{1}{2}m_e v_y^2 + \frac{1}{2}m_e v_z^2$$

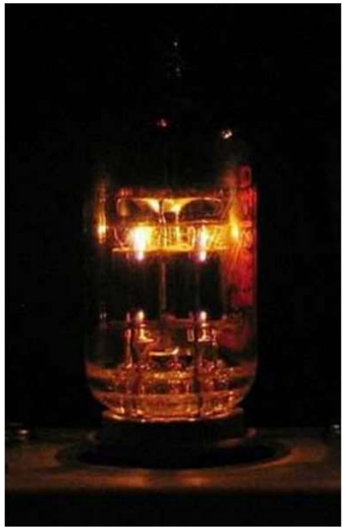
for 'e' emission $\frac{1}{2}mv_x^2 > E_F + \Phi$

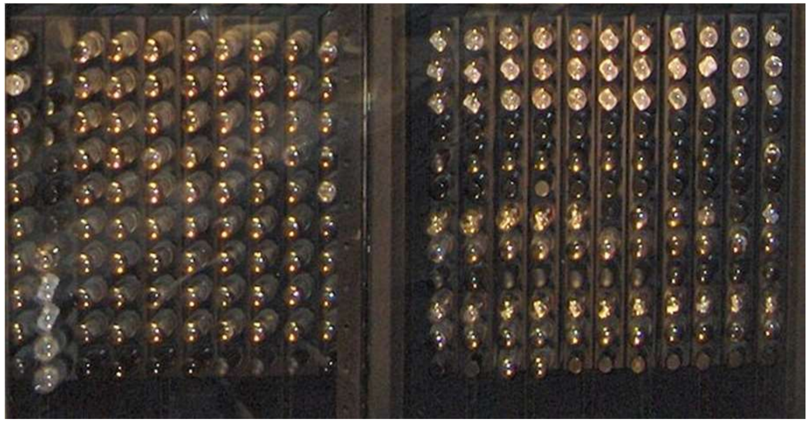


Richardson-Dushman Equation $$J = B_0 T^2 \exp\left(-\frac{\Phi}{kT}\right)$$

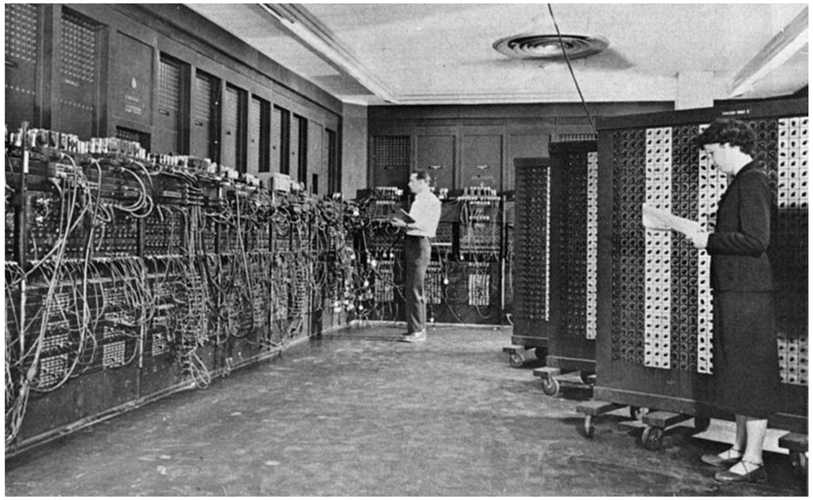
- To obtain an efficient thermionic cathode, we need high temperature and low work functions
- However, high melting temperature materials such as W and Ta, have high work functions, vise versa.
- Combination of high melting temperature materials (base) and low work function materials to make the cathode. E.g., Ni+Th, or W+BaO/ThO/Cao etc.

Vacuum Tubes and ENIAC





ENIAC:1946, 18000 tubes, (2.4m*0.9m*30m), took up 167 m^2, and consumed 150 kW of power.



Thermionic Emission

The Nobel Prize in Physics 1928
Owen Willans Richardson

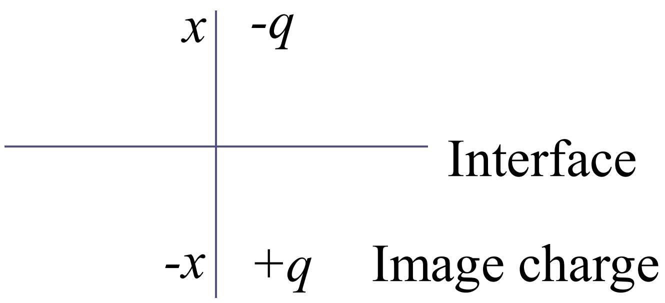
The Nobel Prize in Physics 1928

Owen Willans
Richardson

The Nobel Prize in Physics 1928 was awarded to Owen Willans Richardson *"for his work on the thermionic phenomenon and especially for the discovery of the law named after him"*.

Schottky Effect & Field Emission

Effect of Image Charges

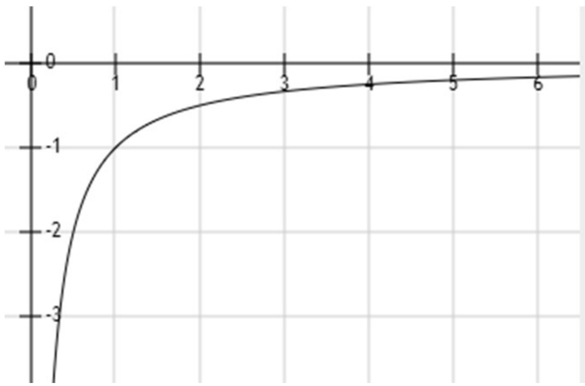


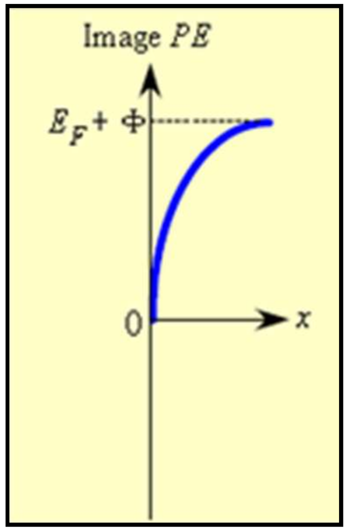
- When a electron moves close to the conductor, a image charge is appeared, hence

$$PE_{image}(x) = -\frac{e^2}{16\pi\varepsilon_0 x}$$

- At x = 0, we define PE = 0. Faraway from the surface, the PE is expected to be $E_F + \Phi$, hence

$$PE_{image}(x) = (E_F + \Phi) - \frac{e^2}{16\pi\varepsilon_0 x}$$





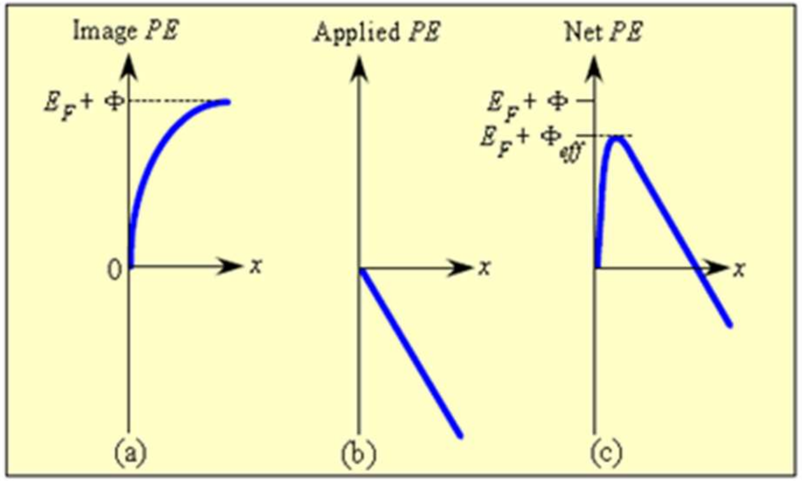
Effect of Applied Filed

- When a voltage difference is applied between the anode and the cathode

$$PE_{applied}(x) = -exE \qquad E: \text{applied field}$$

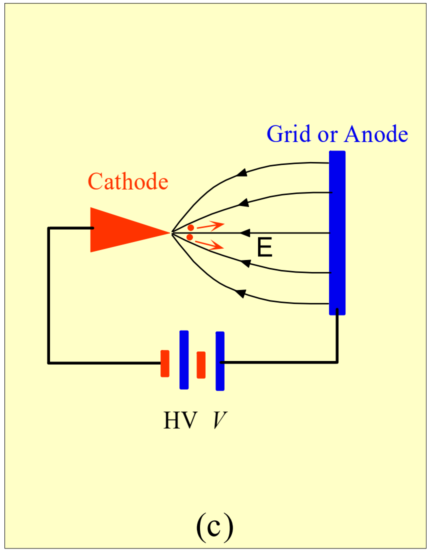
- The total $PE\,(x)$ of the electron outside the metal

$$PE(x) = (E_F + \Phi) - \frac{e^2}{16\pi\varepsilon_0 x} - exE$$



$$\frac{dPE(x)}{dx} = \frac{e^2}{32\pi\varepsilon_0 x^2} - eE = 0 \Rightarrow x = \left(\frac{e}{32\pi\varepsilon_0 E}\right)^{1/2}$$

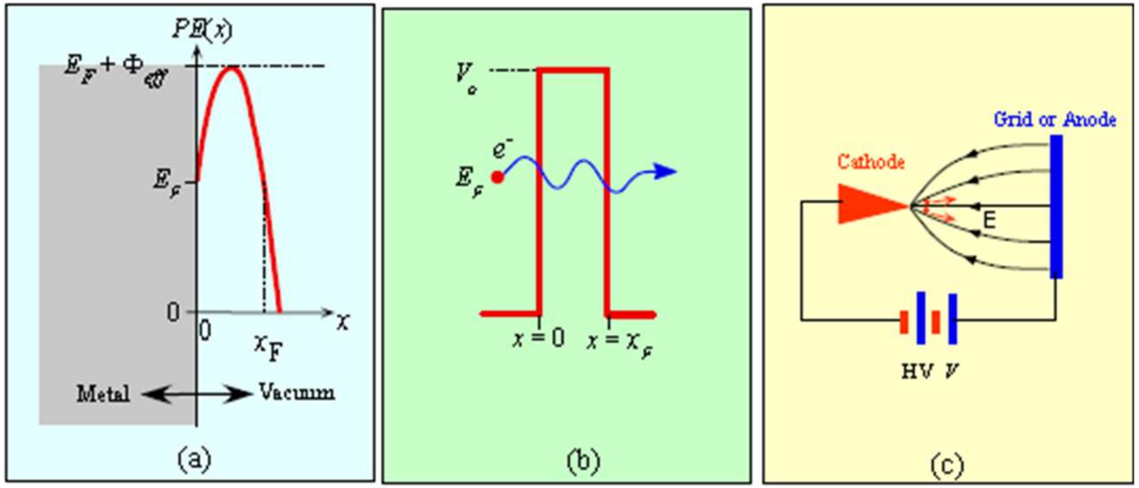
$$PE_{\max} = E_F + \Phi - \left(\frac{e^3 E}{4\pi\varepsilon_0}\right)^{1/2} \qquad \Phi_{effective} = \Phi - \left(\frac{e^3 E}{4\pi\varepsilon_0}\right)^{1/2}$$

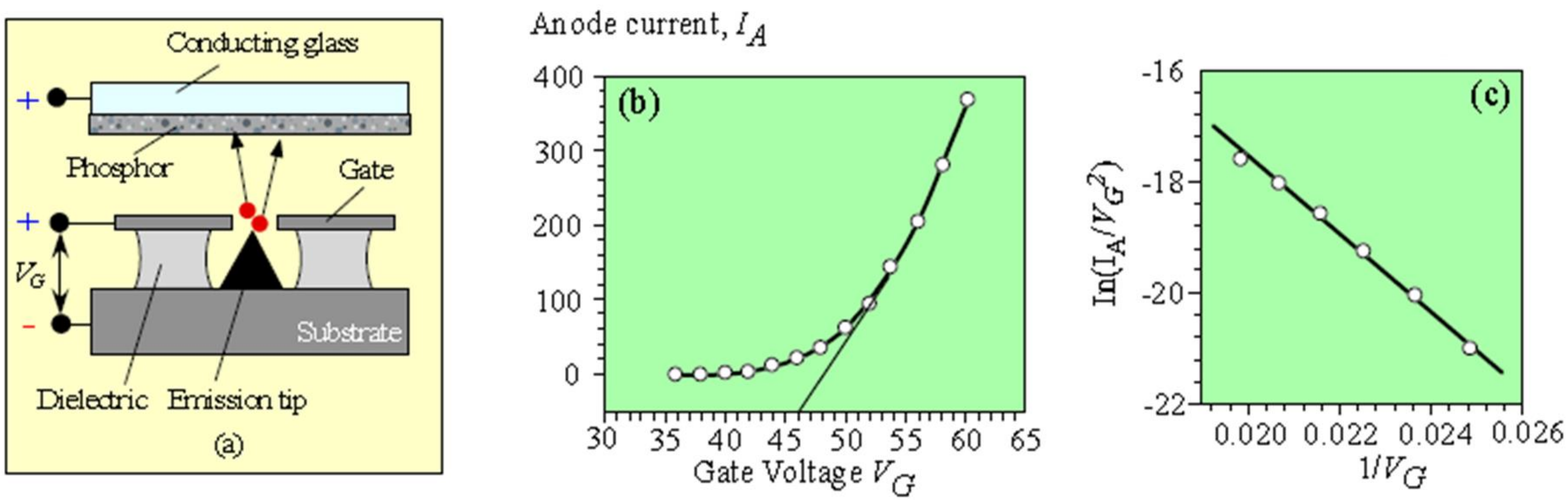


- Schottky Effect: lowing the work function by the applied field

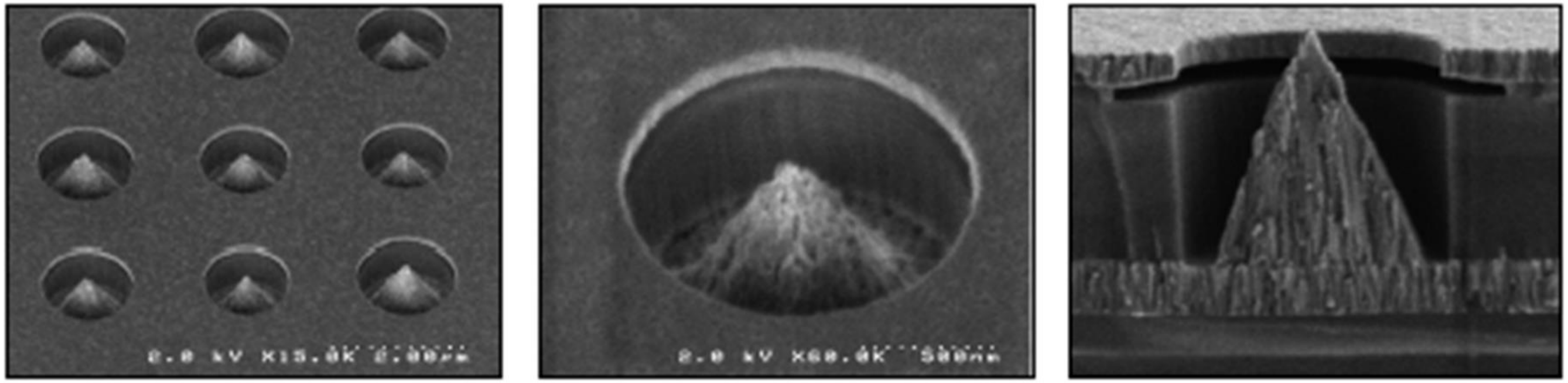
Field Emission

Field emission: When the filed becomes very large, $E > 10^7$ Vcm^{-1}, the $PE\ (x)$ outside the metal surface bend steeply, giving a narrow PE barrier → 'e' at E_F tunneling





Field Emission



Left: A scanning electron microscope image of an array of electron field emitters (icebergs). Center: One iceberg. Right: A cross section of a field emitter. Each iceberg is a source of electron emission arising from Fowler–Nordheim field emission; for further information see B. Chalamala, et al., *IEEE Spectrum*, April 1998, pp. 42–51.

| SOURCE: Courtesy of Dr. Babu Chalamala, Flat Panel Display Division, Motorola.

Field emission has a number of distinct advantages. It is much **more power efficient** than thermionic emission which requires heating the cathode to high temperatures. In principle, field emission can be **operated at high frequencies** (fast switching time) by reducing various capacitances in the emission device or controlling the electron flow with a grid.

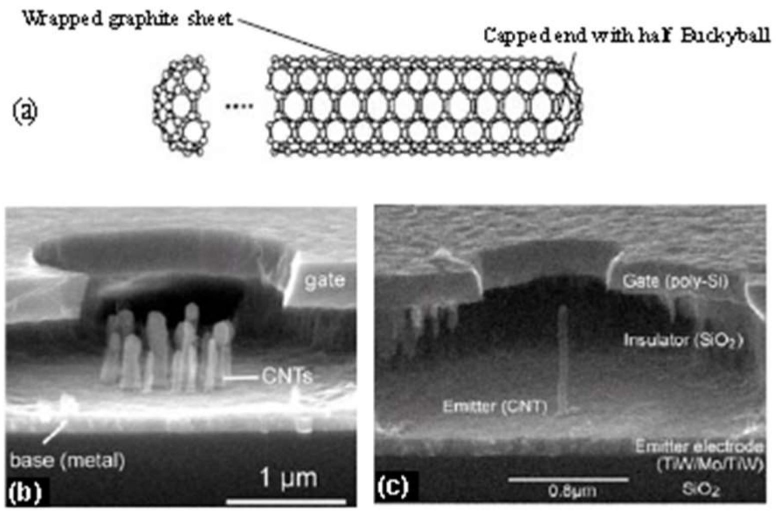
Field Emission

Applications of field emission

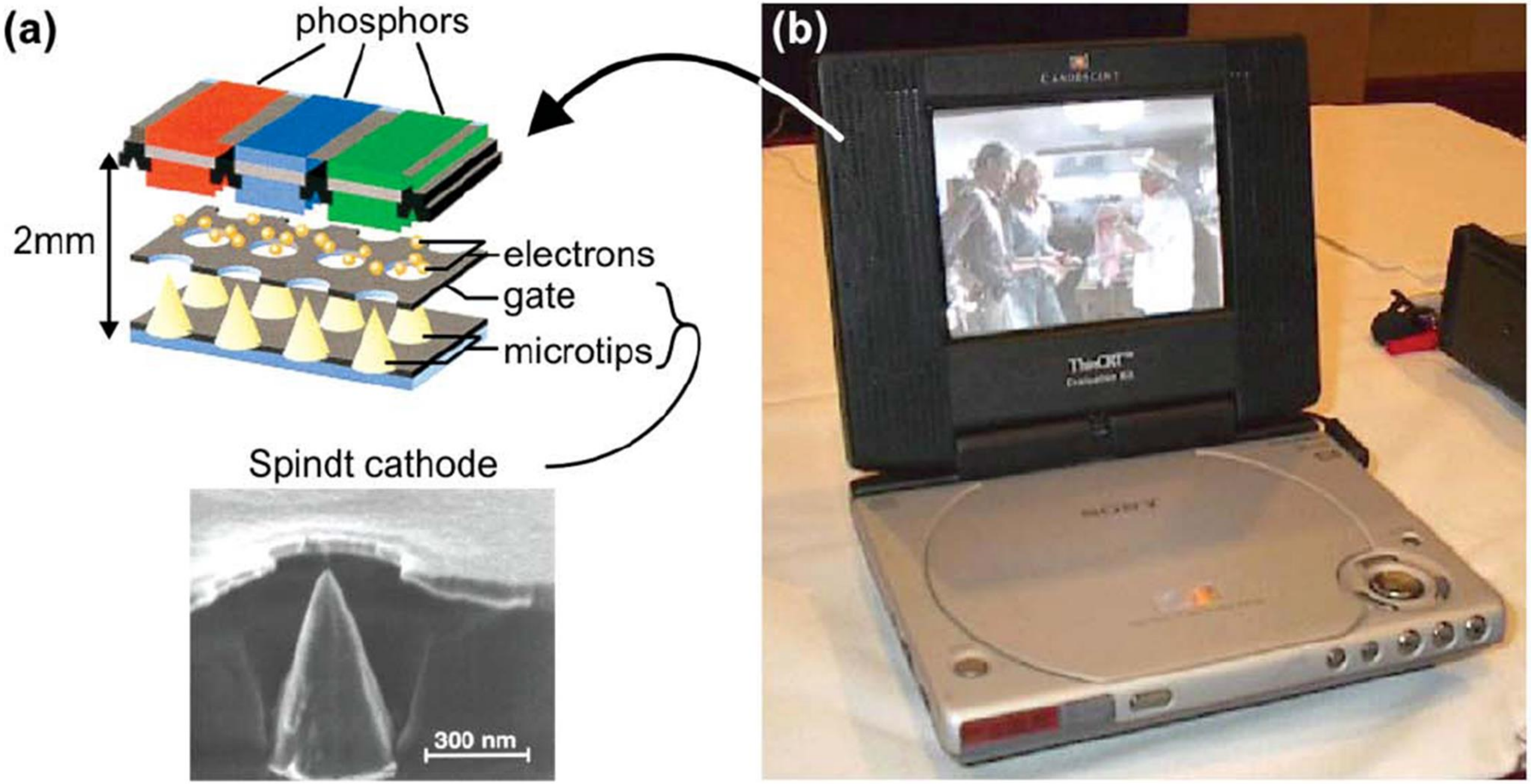
- Field emission microscopy, microwave amplifier (high power and wide bandwidth)
- Parallel electron beam microscopy, nanolithography, portable X-ray generator
- Field emission display (FED): than flat display (~ 2mm thick), a low power consumption, quick start, a wide viewing angle of about 170 degree.

Field emission tip materials

- Mo, W, Hf
- Si (micromachining)
- Carbon nanotube (CNT)
- High aspect ration: diameter ~nanometer rang, length of 10-100 μm; efficient electron emitter

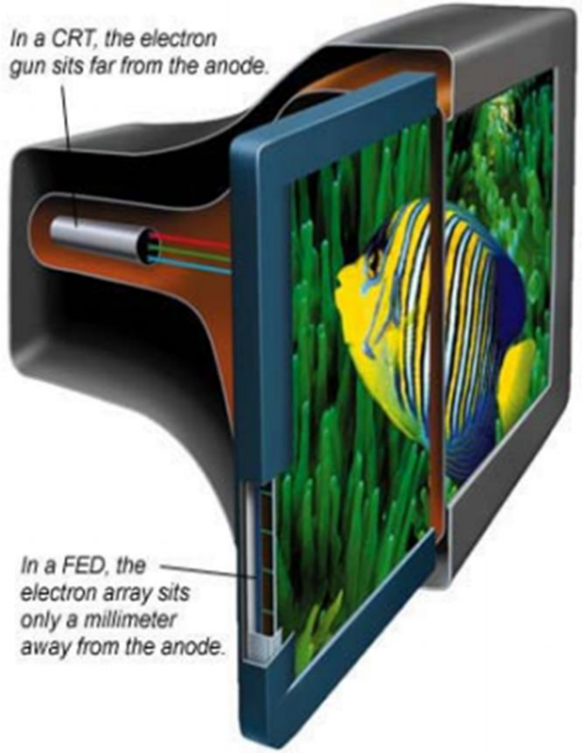


Field Emission Display (FED)

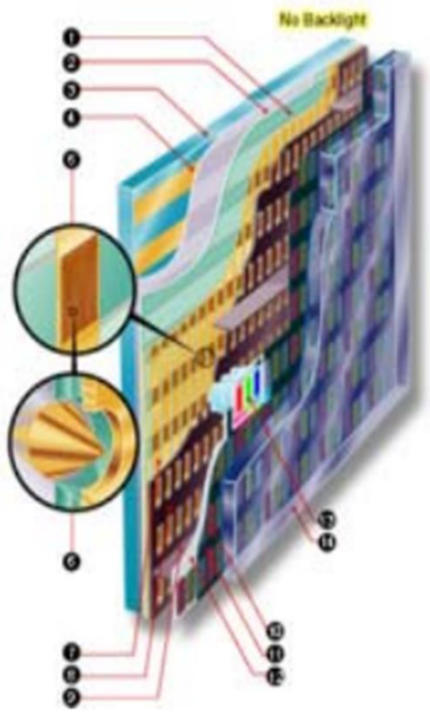


Cross-section of a field emission display showing a Spindt tip cathode, (b) Sony portable DVD player using a field emission display.

Field Emission Display (FED)



Hot cathode



Cold cathode

Cross-Section of a FED

1. Dielectric
2. Patterned Resister Layer
3. Cathode Glass
4. Row Metal
5. Emitter Array
6. Single Emitter Cone & Gate Hole
7. Column Metal
8. Focusing Grid
9. Wall
10. Phosphor
11. Black Matrix
12. Aluminum Layer
13. Pixel On
14. Faceplate Glass

Advantages:
Thinner, lighter, vivid color, fast response, wide viewing angle etc.

Phonon: Harmonic Oscillator

Harmonic Oscillator and Lattice Waves

Quantum Harmonic Oscillator

Harmonic potential energy

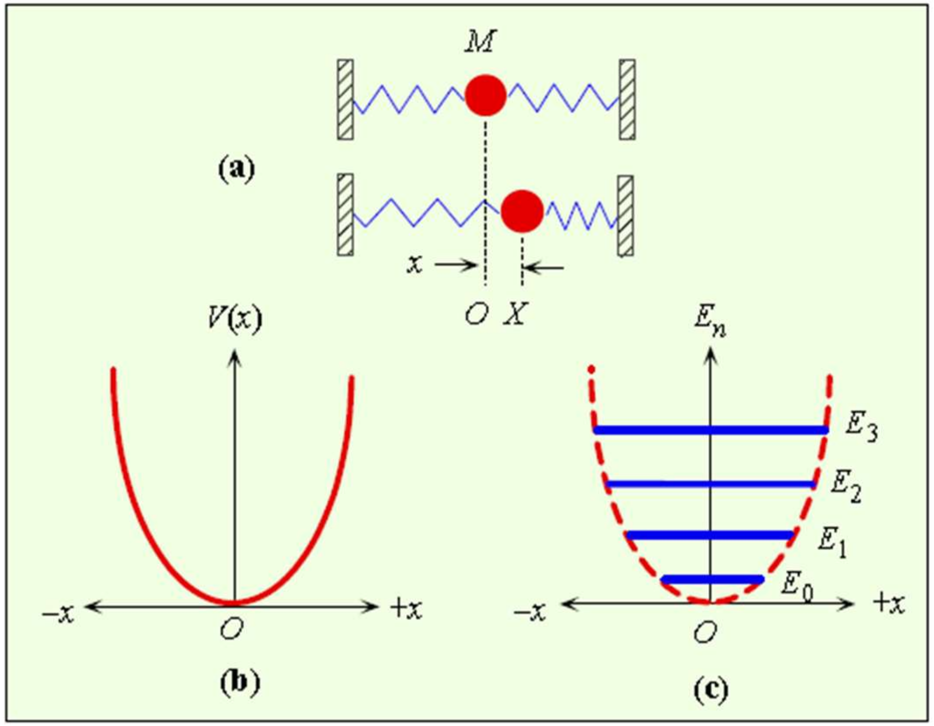
$$V(x)=\frac{1}{2}\beta x^2 \qquad \beta\text{: spring constant}$$

Schrödinger Equation: harmonic oscillator

$$\frac{d^2\varphi}{dx^2}+\frac{2m}{\hbar^2}\left(E-\frac{1}{2}\beta x^2\right)\varphi=0$$

Energy of a harmonic oscillator

$$E_n=\left(n+\frac{1}{2}\right)\hbar\omega \qquad \omega=(\beta/m)^{1/2}$$

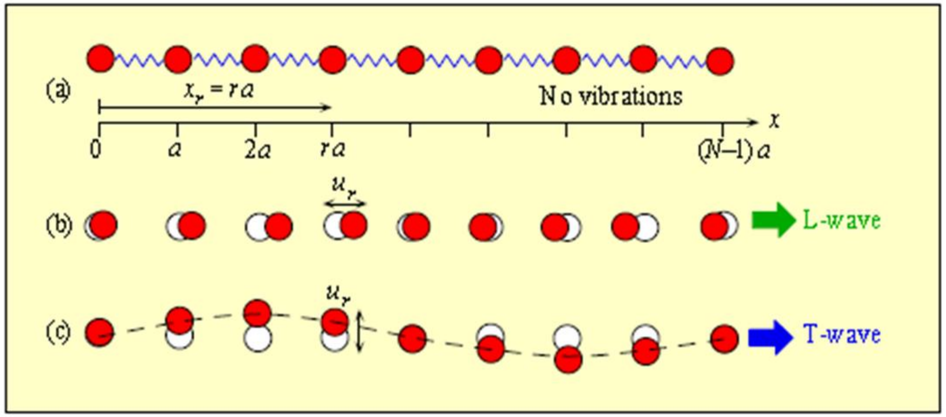


Lattice Wave

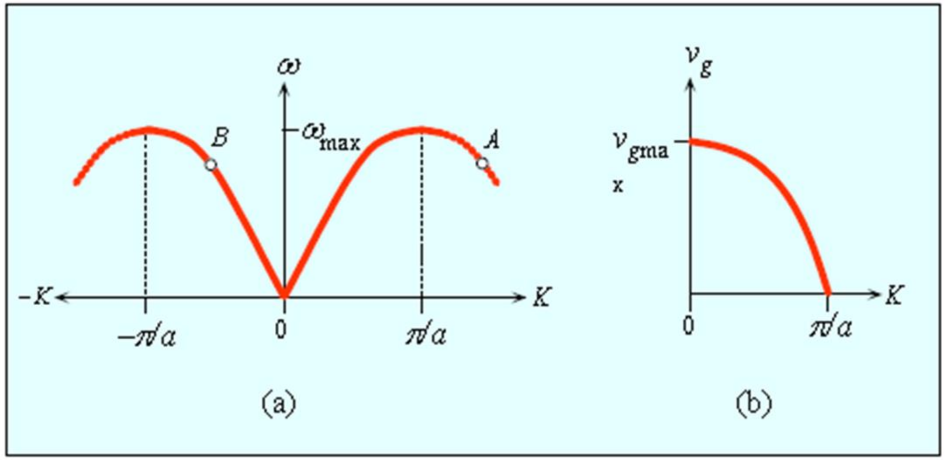
Traveling-wave-type lattice vibration

$$u_r = A\exp\left[j\left(Kx_r - \omega t\right)\right]$$

displacement (u_r), amplitude (A), wave vector (K), equilibrium position (x_r)



- Wavelength of lattice wave $\Lambda = 2\pi/K$
- Phonon energy $E_{phonon} = \hbar\omega = h\nu$
- Phonon momentum $p_{phonon} = \hbar K$
- Dispersion relation $\omega = 2\left(\dfrac{\beta}{M}\right)^{1/2}\left|\sin\left(\dfrac{1}{2}Ka\right)\right|$
- Lattice cut-off frequency $\omega_{\max} = 2\left(\beta/M\right)^{1/2}$



- Group velocity: the velocity which traveling waves carry energy

$$v_g = \frac{d\omega}{dK} = 2\left(\frac{\beta}{M}\right)^{1/2} a\cos\left(\frac{1}{2}Ka\right)$$

Lattice Wave

- Vibrational modes

$$q\frac{\Lambda}{2} = L = Na \qquad q = 1,2,3\ldots$$

$$K = \frac{q\pi}{L} = \frac{q\pi}{Na} \qquad q = 1,2,3\ldots$$

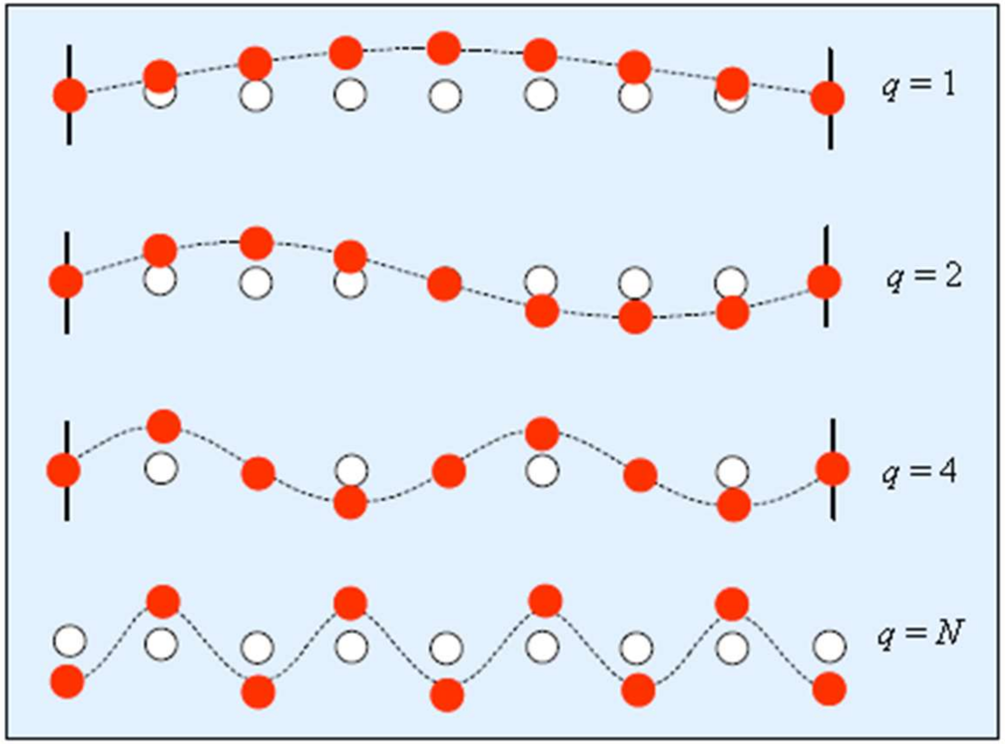
- The smallest wavelength

$$(1/2)\Lambda = a$$

$$K = 2\pi/\Lambda = \pi/a$$

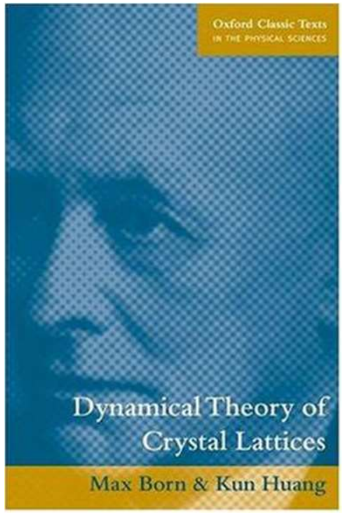
- Lattice vibrational mode in 3-D

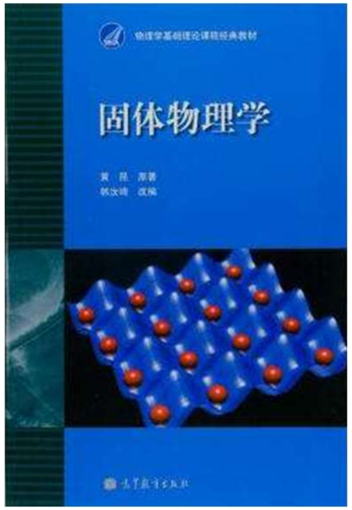
$$K_x = \frac{q_x\pi}{L_x}; \quad K_y = \frac{q_y\pi}{L_y}; \quad K_z = \frac{q_z\pi}{L_z}$$



- Which mode has highest energy?
- Higher mode is excited by temperature
- Number of independent modes is 3N

Solid-State Physics





HUANG Kun
02-Sep-1919 to 06-Jul-2005

Debye Heat Capacity

- Molar heat capacity

$$C_m = dU_m / dT$$

- Average energy of oscillators

$$\overline{E} = \frac{\sum_{n=0}^{\infty} E_n P(E_n)}{\sum_{n=0}^{\infty} P(E_n)} \qquad P(E_n) \propto \exp(-E_n/kT)$$

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega$$

- Average energy of oscillators at ω

$$\overline{E}(\omega) = \frac{\hbar\omega}{\exp\left(\frac{\hbar\omega}{kT}\right) - 1}$$

The phonon concentration in the crystal increases with temperature

- Internal energy of all lattice vibrations

$$\boxed{U_m = \int_0^{\omega_{\max}} \overline{E}(\omega) g(\omega) d\omega}$$

- $g(\omega)$: No. of modes per unit frequency. Density of vibrational states
- $g(\omega)\, d\omega$: No. of state in $d\omega$

- Density of states for lattice vibrations

$$g(\omega) \approx \frac{3V}{2\pi^2} \frac{\omega^2}{v^3}$$

- Debye frequency $\omega_{\max} \approx v\left(6\pi^2 N_A / V\right)^{1/3}$

- Debye temperature $T_D = \dfrac{\hbar\omega_{\max}}{k}$

Above Debye temperature, all vibrational frequencies are executed by the lattice waves

Debye Heat Capacity

Debye Heat capacity: lattice vibrations

$$C_m = 9R\left(\frac{T}{T_D}\right)^3 \int_0^{T_D/T} \frac{x^4 e^x dx}{\left(e^x - 1\right)^2}$$

- Low T: energy increase due to more phonons being created and higher frequencies being excited
- $T>T_D$: energy increase due to more phonons being created

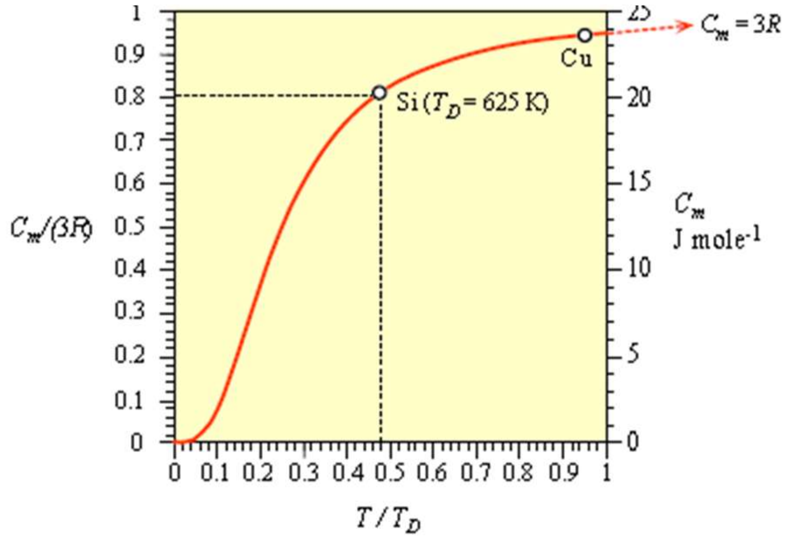


Table 4.5 Debye temperatures T_D, heat capacities, and thermal conductivities of selected elements

	Crystal							
	Ag	Be	Cu	Diamond	Ge	Hg	Si	W
T_D (K)*	215	1000	315	1860	360	100	625	310
C_m (J K^{-1} mol^{-1})†	25.6	16.46	24.5	6.48	23.38	27.68	19.74	24.45
c_s (J K^{-1} g^{-1})†	0.237	1.825	0.385	0.540	0.322	0.138	0.703	0.133
κ (W m^{-1} K^{-1})†	429	183	385	1000	60	8.65	148	173

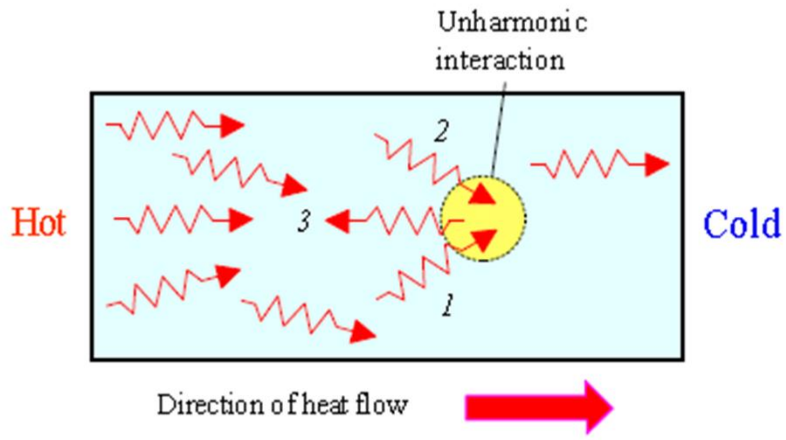
Thermal Conductivity of Nonmetals

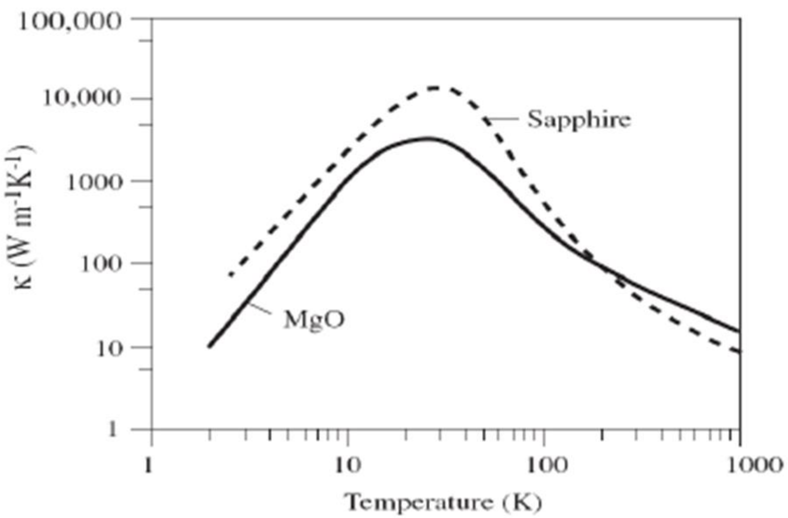
Thermal conductivity due to phonon

$$k = C_{\mathrm{v}} v_{ph} l_{ph}$$

- At $T > T_D$, increase of T:
  - Number of phonons increases
  - Free path decrease due to phonon-phonon collisions
  - Conductivity decrease

- At $T < T_D$, increase of T:
  - C_{v} increase with T^3
  - Number of phonons is small, free path is due to collisions of phonons with crystal imperfections, crystal surface and boundaries.
  - Conductivity increase

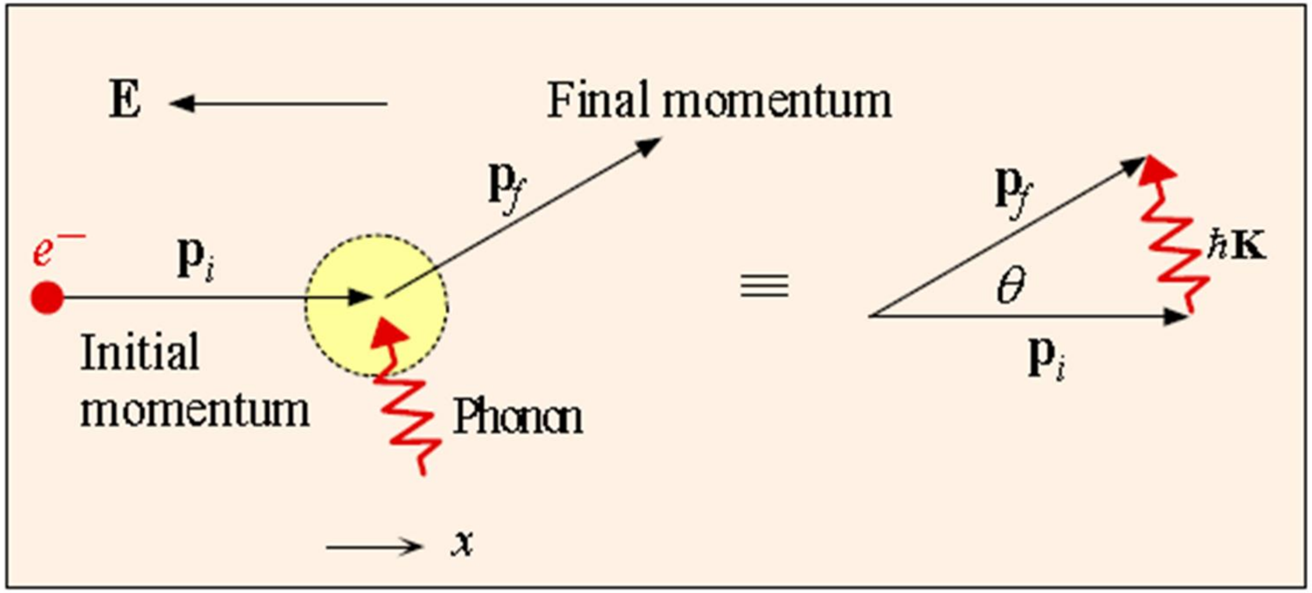




Electrical Conductivity

- Electrical conductivity $T > T_D$ $\qquad \sigma \propto \tau \propto \frac{2}{n_{ph}} \propto \frac{1}{T}$

- Electrical conductivity $T < T_D$ $\qquad N = \frac{\left(2m_e v_F\right)^2}{\left(\hbar K\right)^2} \propto \frac{1}{T^2} \qquad \sigma \propto N_\tau \propto \frac{N}{n_{ph}} \propto \frac{1}{T^5}$



Low angel scattering of a conduction electron by a phonon